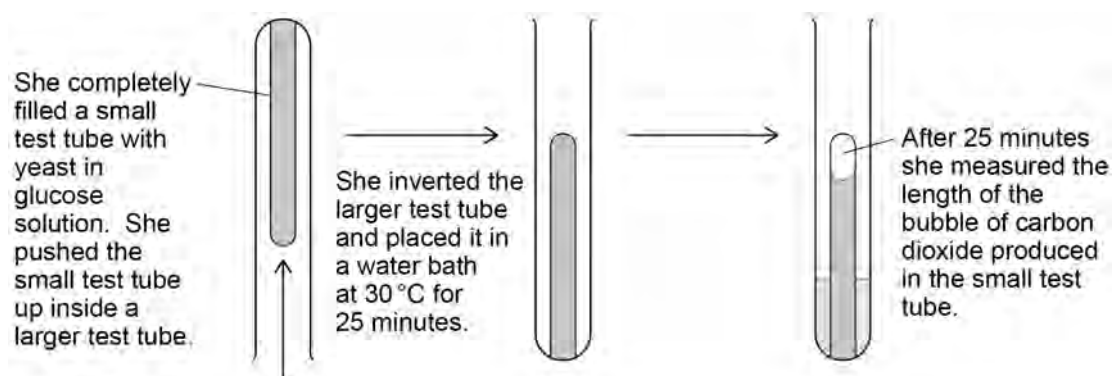


0 2

A student investigated the effect of different sugars on the rate of respiration in yeast. Yeast normally respires glucose.

Figure 2 shows the method she used for her first experiment.

Figure 2



0 2

1

Other than those stated, suggest **two** variables the student needed to keep constant in her investigation.

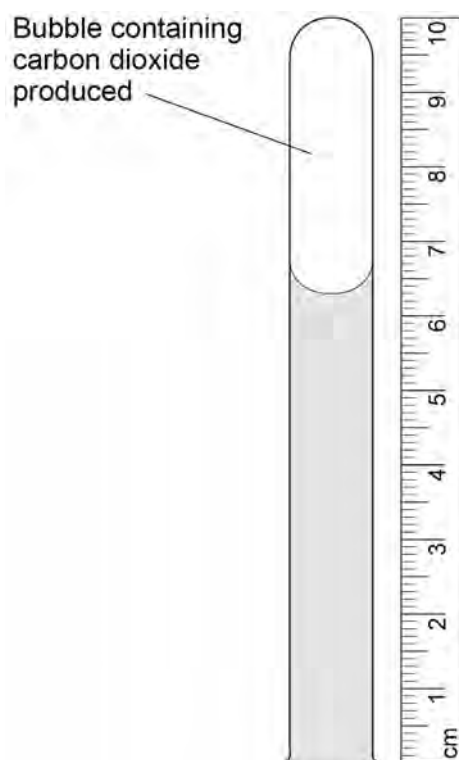
[1 mark]

1 _____

2 _____

Figure 3 shows the result she obtained for yeast in glucose solution.

Figure 3



0 2 . 2

Use **Figure 2** and **Figure 3** to calculate the rate of carbon dioxide production in mm s^{-1} for yeast in glucose solution.

Give your answer in standard form **and** to 2 significant figures.

Show your working.

[2 marks]

Answer _____ mm s^{-1}

Turn over ►

0 2 . 3

The student repeated the experiment using yeast in maltose solution. She found the rate of carbon dioxide production was slower than with yeast in glucose solution. Suggest why.

[2 marks]

0 2 . 4

A second student used a different method to investigate the effect of different sugars on the rate of respiration in yeast.

He set up a tube with yeast in glucose solution and added bromothymol blue. Bromothymol blue changes from blue to yellow when carbon dioxide is produced.

To determine the rate of respiration, he timed how long it took for the solution to change from blue to yellow.

Suggest:

- why the method the second student used would be less accurate than the method the first student used
- how the accuracy of the method the second student used could be improved.

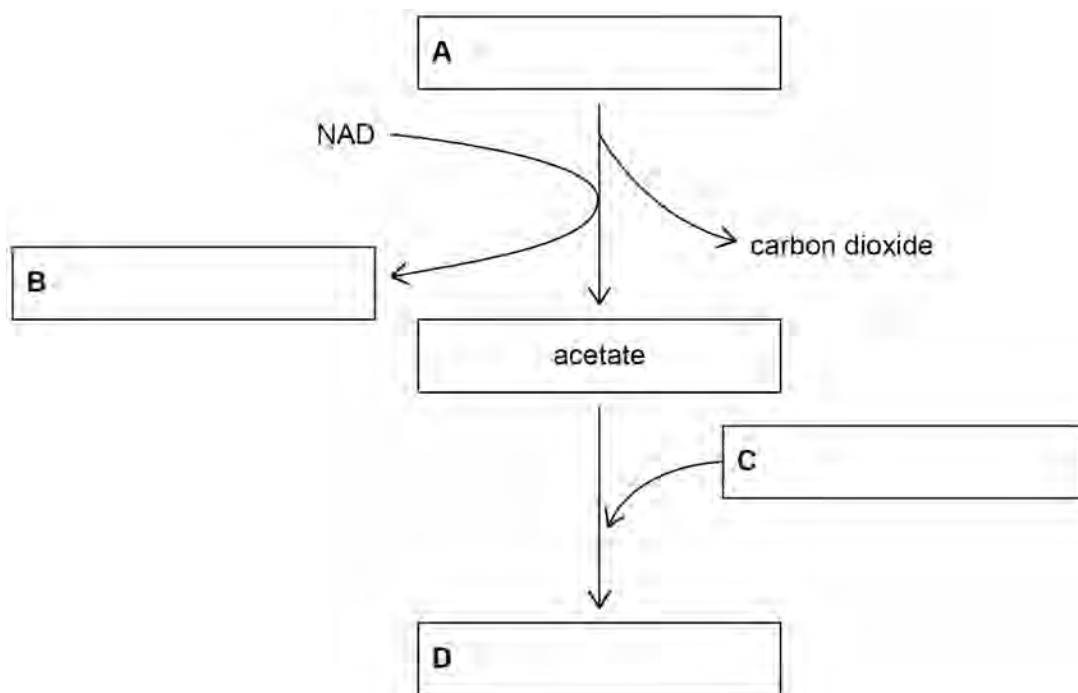
[2 marks]

This method would be less accurate because

The accuracy of this method could be improved by

0 2 . 5 Complete the boxes **A** to **D** in **Figure 4** to show the link reaction.

Figure 4



[2 marks]

Turn over for the next question

Turn over ►

